Towards Foundation Models for Shock Compression of Solid Reactive Materials

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Condensed phase reactive materials (propellants, explosives, pyrotechnics, structural reactive materials, etc.) exhibit strongly coupled thermo-chemical and mechanical responses that depend on both chemical composition and microstructural configuration. Although high-fidelity direct numerical simulations (DNS) can resolve these interactions, their extreme computational cost severely limits exploration of chemical space, restricting systematic understanding of chemistry–structure–performance relationships. Neural surrogate models have been introduced to alleviate this burden; however, existing approaches typically focus on varying microstructure within a fixed chemistry or consider multiple chemistries without explicitly resolving geometric variability, limiting their ability to generalize across coupled chemistry–structure regimes. Taking a step toward foundation models for two-component energetic materials (EMs, e.g., crystals and defects), in this work we introduce OmniPARC, a physics-aware neural surrogate model designed to operate across chemical spaces and varying microstructural geometries. The model is conditioned on chemistry and structured to share common physical behavior across systems while allowing chemistry-specific responses to emerge through modulation of the learned dynamics, enabling generalization across different chemistries while maintaining physically consistent evolution. OmniPARC is trained on a small set of five representative EM CHNO species and demonstrates strong performance across these chemically distinct systems. Importantly, the model extends beyond the training chemistries, exhibiting reasonable predictive capability on unseen species while capturing both shared physical trends and chemistry-specific deviations in the underlying dynamics. Beyond prediction, OmniPARC enables systematic interrogation of chemistry-structure-performance relationships, including sensitivity analyses across chemistry and microstructure, as well as comparative assessment of self-detonation behavior across chemical species with varying microstructural geometries. This work represents a step toward foundation model for compressed reactive material mixtures, demonstrating how a single neural surrogate can support generalizable and systematic analysis across chemical spaces and microstructural configurations. By bridging chemistry, structure, and dynamics within a unified model, OmniPARC provides a practical pathway for extracting physical insight from limited high-fidelity simulations and constitutes a step toward foundation model for reactive materials.